ClimateAdapt Pune: Evidence-Led Climate Action Report

Pilot Study: Pune Urban Core, Maharashtra | Generated: 18 October 2026 04:59 UTC

Executive Summary: This report translates real municipal drainage statistics (OpenCity Pune), OpenStreetMap infrastructure assets (5,779 trees, 267 medical facilities, waterways), and Open-Meteo 7-day atmospheric forecasts into actionable climate resilience priorities for Pune's electoral wards. In strict adherence to the project charter, all findings distinguish measured infrastructure data from model proxies, and unvalidated data (such as local well hydrographs or cloud-covered satellite rasters) are explicitly labeled as missing rather than assumed.

1. High-Priority Wards (Screening Indicators)

Scores are normalized 0-100 screening proxies. Higher values denote greater exposure or infrastructure deficit.

Ward / Prabhag	Heat Proxy	Flood Proxy	Water Stress	Green Deficit	Composite
Vadgaon Sheri - Ganesh N	58.4	75.2	52.0	95.0	68.8
Arunkumar Vaidya Stadium	64.4	54.8	60.0	95.0	65.0
Ghorpade Garden	63.8	54.6	60.0	93.3	64.4
Upper Indiranagar	58.4	59.1	60.0	95.0	64.4
Mohammadwadi	58.4	60.7	52.0	95.0	63.7
Vimannagar Sanjay Park	58.4	55.4	60.0	95.0	63.1

2. Prioritized Adaptation Action Plan (Budget Allocation)

Target Portfolio Budget: INR 5,000,000 | **Committed Allocation:** INR 5,000,000 (100.0% efficiency)

Intervention	Category	Units	Cost (INR)	Dept	Lead Time
Shaded Bus Stops & Public Co	Heat & Urban Co	11 station	■1,980,000	PMPML & PMC Health	2 mo
Cool Roof Reflective Coating	Heat & Urban Co	15 building	■1,800,000	PMC Building & Dev	1 mo
Urban Shade Corridors & Nati	Heat & Urban Co	4 km corri	■1,000,000	PMC Garden Departm	3 mo
Decentralized Rainwater Harv	Water Security	1 building	■220,000	PMC Water Supply D	2 mo

3. Climate Stress Scenario Comparisons

Scenario	Heat Index	Flood Index	Water Stress	Composite Index
Current Baseline	53.4	48.8	44.0	52.5
Extreme Heatwave (+3.5°C)	73.4	48.8	59.0	61.8
Extreme Monsoon Cloudburst	43.4	73.8	34.0	59.8
Post-Adaptation Plan	30.0	48.8	42.2	44.0

4. Responsible Use & Data Quality Disclosures

- ***Distinction of Evidence:*** *Drainage statistics are verified 2015-2016 OpenCity PMC releases. Atmospheric values represent Open-Meteo multi-variable NWP forecast models and must not be cited as permanent climate normals.*
- ***Explicit Missing Data:*** *Local observation wells (GSDA telemetry) and uncalibrated DEM hydrological runoff grids remain unconfirmed and are strictly excluded from automated physical simulation to prevent false certainty.*
- ***Human Review:*** *Action allocations are exploratory decision-support calculations and require municipal site surveys before civil construction.*